

Recap lecture 23

Mealy machines in terms of sequential circuit

Regular languages

As already been discussed earlier that any language that can be expressed by a RE is said to be regular language, so if L_1 and L_2 are regular languages then $L_1 + L_2$, L_1L_2 and L_1^* are also regular languages. This fact can be proved by the following two methods

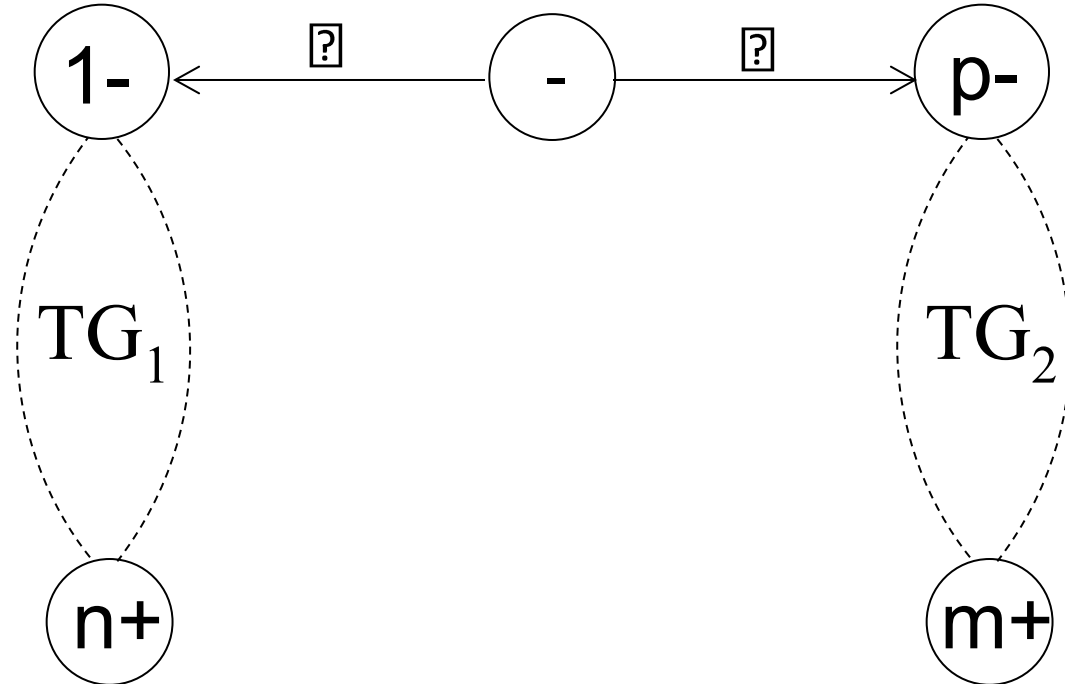
1) **By Regular Expressions**: As discussed earlier that if r_1, r_2 are regular expressions, corresponding to the languages L_1 and L_2 then the languages $L_1 + L_2$, L_1L_2 and L_1^* generated by $r_1 + r_2$, r_1r_2 and r_1^* , are also regular languages.

Regular languages continued ...

2) **By TGs**: If L_1 and L_2 are regular languages then L_1 and L_2 can also be expressed by some REs as well and hence using Kleene's theorem, L_1 and L_2 can also be expressed by some TGs. Following are the methods showing that there exist TGs corresponding to $L_1 + L_2$, L_1L_2 and L_1^*

Regular languages continued ...

If L_1 and L_2 are expressed by TG_1 and TG_2 then following may be a TG accepting $L_1 + L_2$



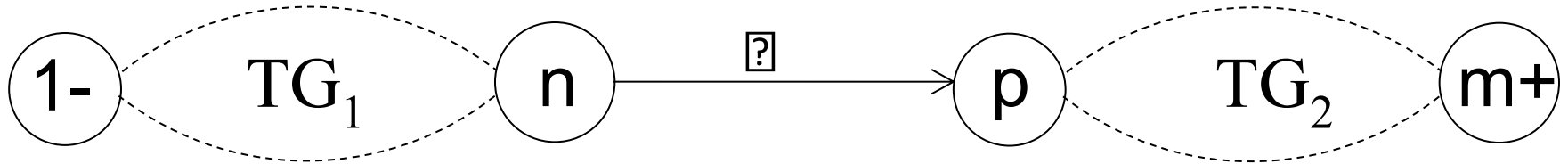
Regular languages continued ...

If L_1 and L_2 are expressed by the following TG_1 and TG_2

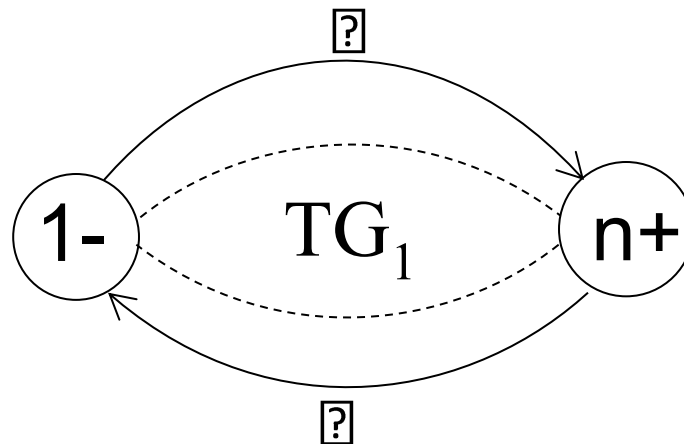


then following may be a TG accepting L_1L_2

Regular languages continued ...

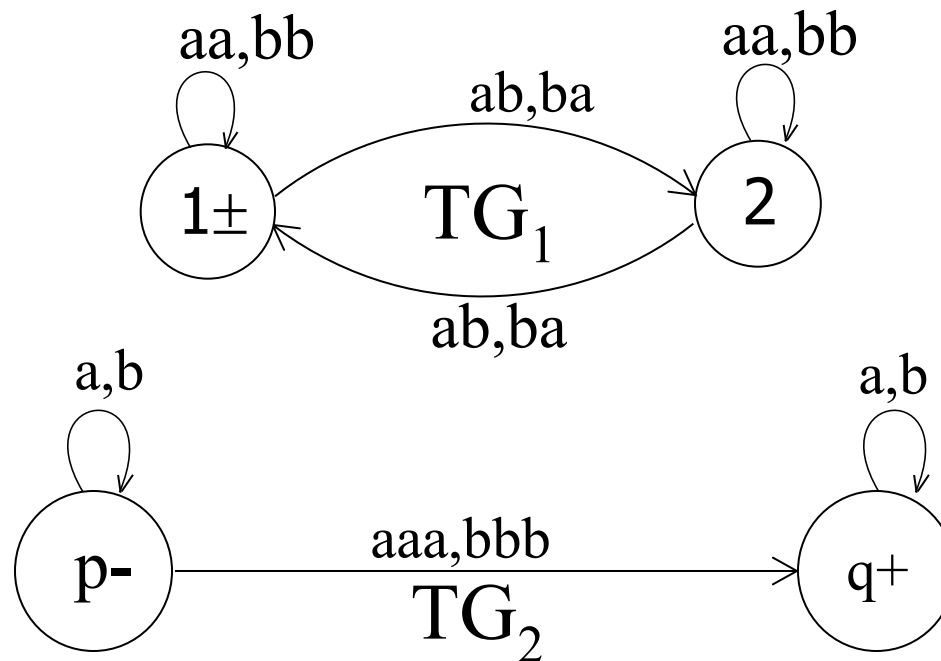


also a TG accepting L_1^* may be as under



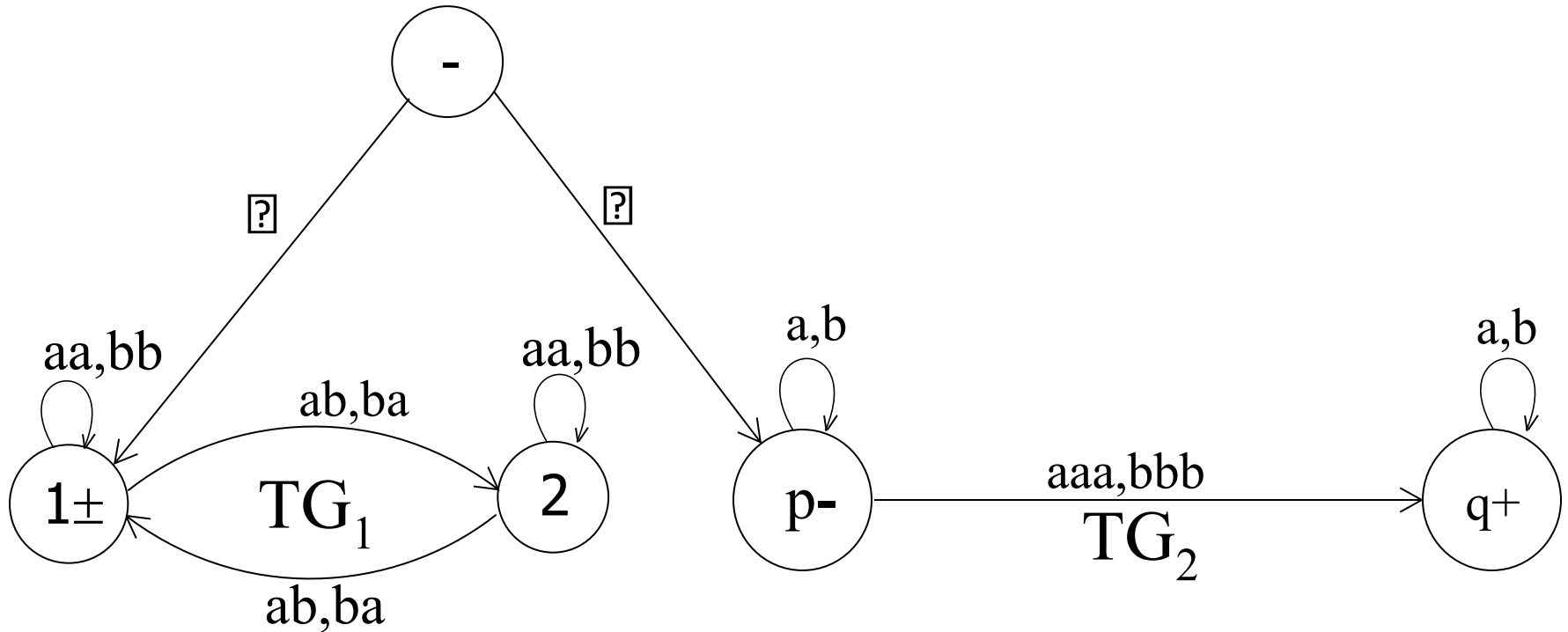
Example

Consider the following TGs



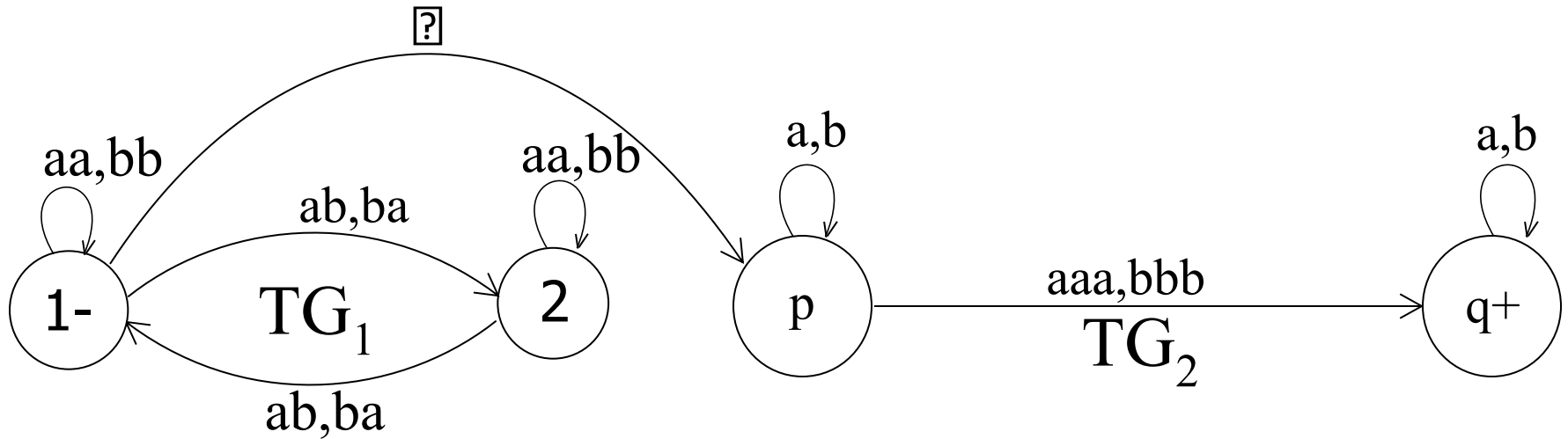
Following may be a TG accepting $L_1 + L_2$

Example continued ...



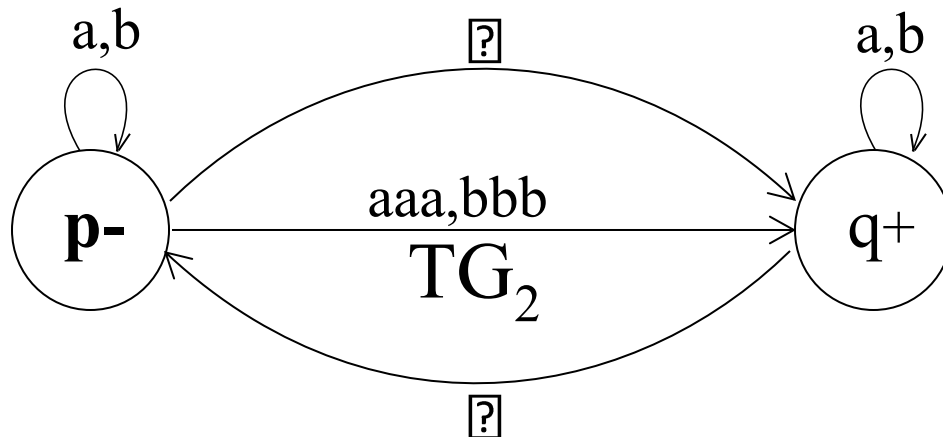
also a TG accepting L_1L_2 may be

Example continued ...



and a TG accepting L_2^*

Example continued ...



Complement of a language

Let L be a language defined over an alphabet Σ , then the language of strings, defined over Σ , **not belonging to L** , is called **Complement of the language L** , denoted by L^c or $\bar{L}$.

Note: To describe the complement of a language, it is very important to describe the alphabet of that language over which the language is defined.

For a certain language L , the complement of L^c is the given language L *i.e.* $(L^c)^c = L$

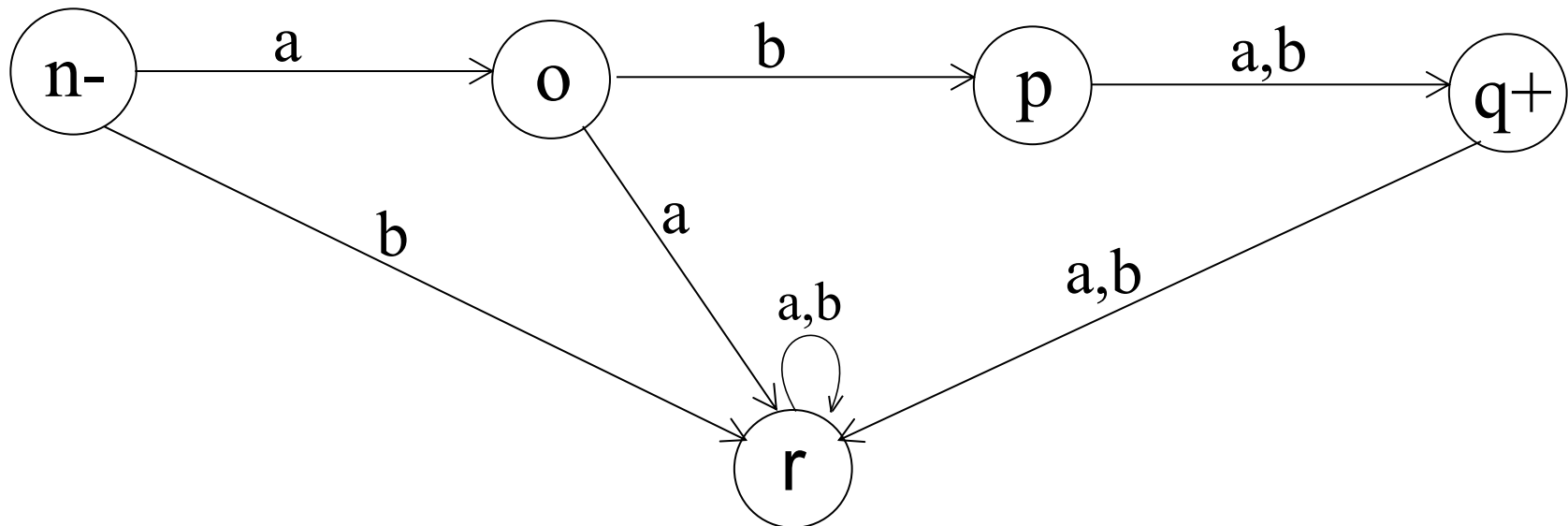
Theorem: If L is a regular language then, L^c is also a regular language.

Proof of the theorem

Proof: since L is a regular language, so by Kleene's theorem, there exists an FA, say F , accepting the language L . Converting each of the final states of F to non-final states and old non-final states of F to final states. FA, thus, obtained will reject every string belonging to L and will accept every string, defined over Σ , not belonging to L . Which shows that the new FA accepts the language L^c . Hence using Kleene's theorem L^c can be expressed by some RE. Thus L^c is regular.

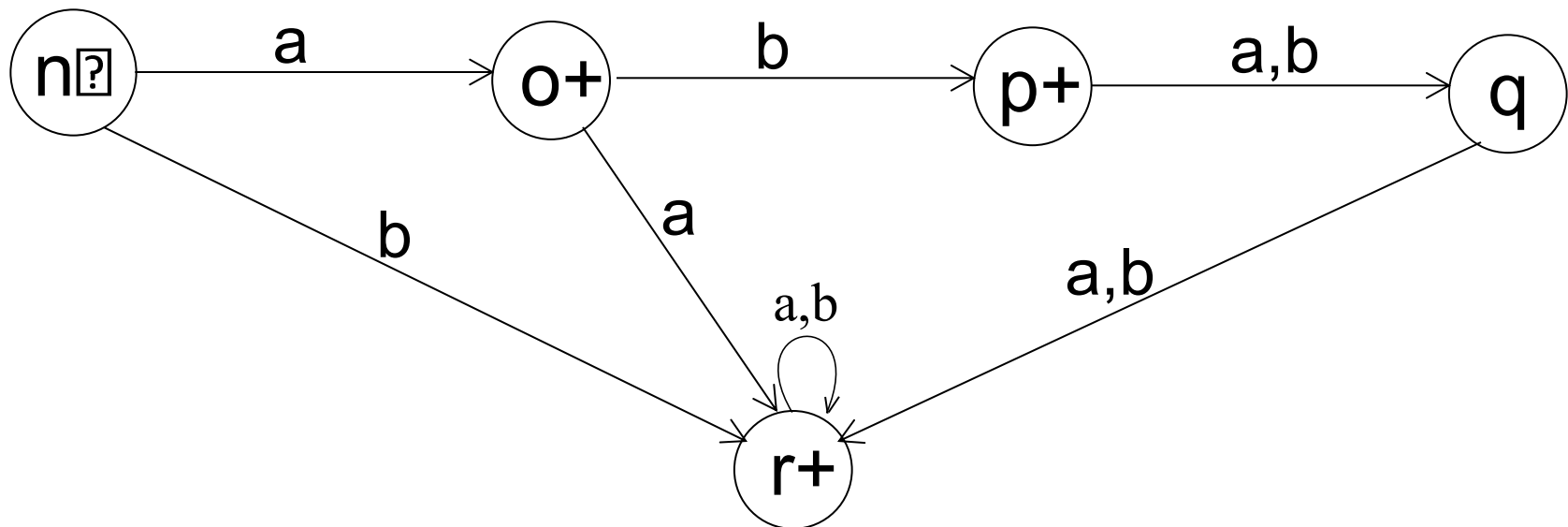
Example

Let L be the language over the alphabet $\Sigma = \{a, b\}$, consisting of only two words aba and abb , then the FA accepting L may be



Example continued ...

Converting final states to non-final states
and old non-final states to final states,
then FA accepting L^c may be



Theorem

Statement: If L_1 and L_2 are two regular languages, then $L_1 \cap L_2$ is also regular.

Proof: Using De-Morgan's law for sets

$$(L_1^c \cup L_2^c)^c = (L_1^c)^c \cap (L_2^c)^c = L_1 \cap L_2$$

Since L_1 and L_2 are regular languages, so are L_1^c and L_2^c . L_1^c and L_2^c being regular provide that $L_1^c \cup L_2^c$ is also regular language and so $(L_1^c \cup L_2^c)^c = L_1 \cap L_2$, being complement of regular language is regular language.

Summing Up

Regular languages, complement of a language, theorem, proof, example, intersection of two regular languages